

Secure 100-Gbit/s Real-time Spatially Multiplexed Quantum Random Number Generation Based on Vacuum Fluctuations

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Abstract We demonstrate high-speed unpredictable random number generation using a single laser diode and parallel homodyne detectors. A spatially multiplexed setup with eight channels mitigates the trade-off between security and generation rate and achieves real-time 100-Gbit/s operation with a low security parameter of 5×10^{-30} . ©2026 The Author(s)

Introduction

Random numbers with genuine unpredictability play a crucial role in modern communication and information-processing systems, particularly in security-sensitive applications. Unlike pseudorandom numbers produced by deterministic algorithms, such randomness must be derived from irreversible physical processes. To this end, physical random number generators exploiting natural random phenomena have been widely investigated. Among them, quantum random number generators (QRNGs) attract particular attention, as certain quantum phenomena, such as vacuum fluctuations, exhibit fundamental unpredictability rooted in quantum mechanics [1,2]. Owing to this property, QRNGs have been adopted in a variety of security-sensitive applications, including quantum key distribution [3], physical-layer encryption [4], and security enhancement of public key infrastructures [5].

QRNGs with output rates ranging from several hundred Mbit/s to around 1 Gbit/s have been commercialized [6,7]. Recently, high-speed QRNGs based on vacuum fluctuation have been actively studied. The QRNG measures vacuum fluctuations using a balanced homodyne detector (BPD) with a coherent local oscillator (LO) [8]. High-speed real-time random number generation at 18.8 Gbit/s was demonstrated using a wideband BPD [9]. A spatial-multiplexing technology was introduced to achieve a higher rate with parallel operation. Offline setups demonstrated a high aggregated generation rate of 100 Gbit/s or higher [10,11]. Real-time spatially multiplexed QRNG with post-processing implemented in an FPGA achieved 50 Gbit/s operation with four channels [12] and 64.9 Gbit/s operations with two channels [13]. One of the technical challenges for real-time operation at higher rates is to maintain high security, i.e., a high degree of unpredictability in the output random numbers. A recent preliminary report on real-time operation with two channels indicated that a security parameter of hashing ϵ_{hash} was increased to be 6.39×10^{-8} at a rate of 100 Gbit/s [14]. Such a high value prevents long-term continuous secure operation with a low failure probability.

This paper first presents 100-Gbit/s real-time QRNG maintaining high security with a very low security parameter ϵ_{hash} of less than 5×10^{-30} . The security parameter is determined mainly based on the power ratio of quantum noise to additive classical noise (noise clearance) in the balanced photodetection and the matrix size in the hashing process. Wideband photodetection typically induces more classical noise and reduces the clearance, and the security decreases. A large matrix size improves the security; however, real-time implementation requires larger amount of FPGA resource as the processing speed per channel increases. Thus, increasing the number of parallel channels while keeping the speed per channel at a moderate level, the so-called ‘slow and wide’ strategy, is an effective approach for realizing higher-speed QRNGs. Here we demonstrate a real-time eight-channel spatially multiplexed QRNG operating at 100 Gbit/s. A low security parameter of less than 5×10^{-30} is achieved in a stable manner. The setup achieves highly secure continuous operation for 100 years with a low failure probability of 3.1×10^{-13} .

Operating Principles

A typical QRNG based on vacuum fluctuation uses a simple half beam splitter for balanced homodyne detection. A high-rate operation requires a wideband BPD and a high-speed analog-to-digital converter (ADC). However, such wideband components introduce additive classical noise, and the security of the QRNG, typically discussed using conditional min-entropy and security parameter analysis [15], decreases. We employed a spatial multiplexing technology using an optical splitter to achieve a high random number generation rate while maintaining high security. Fig. 1 shows the configuration. The spatial multiplexing setup uses a $1 \times 2N$ splitter for N -channel parallel operation. The operating bandwidth of each channel is set at a moderate value ($\sim$ GHz), and the increase in the channel number N enables a high aggregated generation rate. The clearance from classical noise, defined as the power ratio of quantum noise to classical noises, maintained at a high level. The high clearance leads to a high

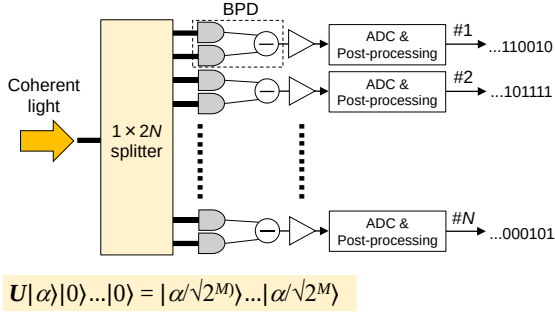


Fig. 1: Configuration of a spatially multiplexed QRNG based on vacuum fluctuations.

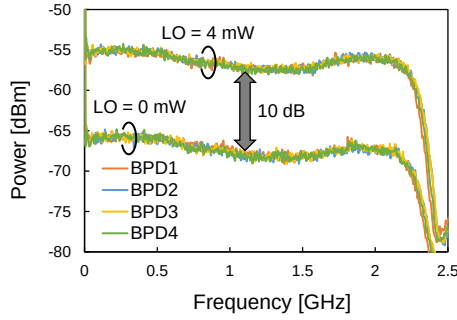


Fig. 2: Electrical spectra of vacuum fluctuations obtained using a four-channel wideband BPD circuit board for LO power of 0 and 4 mW.

conditional min-entropy, enabling a lower security parameter and high security. Furthermore, a larger hashing matrix can be used in the post-processing because the processing speed for each channel is moderate. Thus, the setup mitigates trade-off between the security and generation rate in the QRNG.

The number of multiplexed channels N was limited to four in the previous 50-Gbit/s operation [12] because relatively high LO power of 20 mW for each channel was required to achieve a clearance of approximately 10 dB. We recently developed a four-channel BPD circuit board for a tabletop 50-Gbit/s QRNG prototype [16]. Fig. 2 shows the spectra of vacuum fluctuations when the circuit board was used for detection. The LO power for each channel was set at 0 and 4 mW. Careful circuit design achieved a clearance of approximately 10 dB across more than 2 GHz. The power reduction from 20 to 4 mW indicates that N can be further increased for a higher generation rate while maintaining high clearance.

Experiments

We demonstrate real-time 100-Gbit/s random number generation with eight channels ($N = 8$). Fig. 3 shows the experimental setup. A distributed feedback (DFB) laser diode operating at $1.31 \mu\text{m}$ was used as an LO source. A 1×16 integrated optical splitter split the laser light for eight-channel parallel operation. The power imbalances between two beams for balanced detection were suppressed below 0.15 dB. Two sets

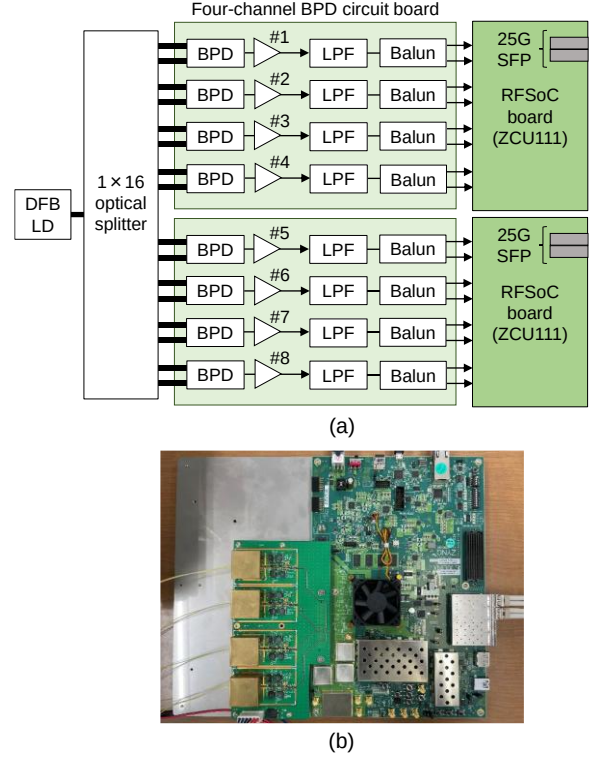


Fig. 3: Experimental setup of 100-Gbit/s (12.5 Gbit/s $\times$ 8 channels) quantum random number generation: (a) schematic diagram and (b) photograph of the four-channel BPD circuit board connected to the RFSoc evaluation board.

of four-channel BPD circuit boards and RFSoc evaluation boards (Zynq UltraScale+ RFSoc ZCU111) were prepared, enabling the simultaneous detection of vacuum fluctuations with eight channels. The RFSoc board consists of 4-Gsample/s ADCs with 12-bit resolution and FPGA. Down-sampling from 4 to 2 Gsample/s, amplitude adjustment that reduces the number of bits per sample from 12 to 10 bits and hashing with a compression ratio η of 0.625 were implemented real time in the FPGA [12]. The hashing employed a 1024×640 Toeplitz matrix with random seed bits. Each channel generated random numbers at $2 \text{ Gsample/s} \times 10 \text{ bits} \times 0.625 = 12.5 \text{ Gbit/s}$. The eight-channel parallel operation achieved an aggregated generation rate of 100 Gbit/s. The 100-Gbit/s random numbers were digitally output via four 25G SFP modules. Truly continuous 100-Gbit/s real-time operation was demonstrated. Four 25-Gbit/s random number outputs were received and stored on a personal computer for evaluation.

First, we measured the noise variances for different LO power by using histogram monitoring implemented in the FPGA. Fig. 4(a) shows the results. The horizontal axis indicates the total LO power launched into the splitter for the eight-channel operation. The small variation between channels is mainly attributed to the variation of optical splitting ratio and slightly different efficiencies of BPDs. The noise variances of each

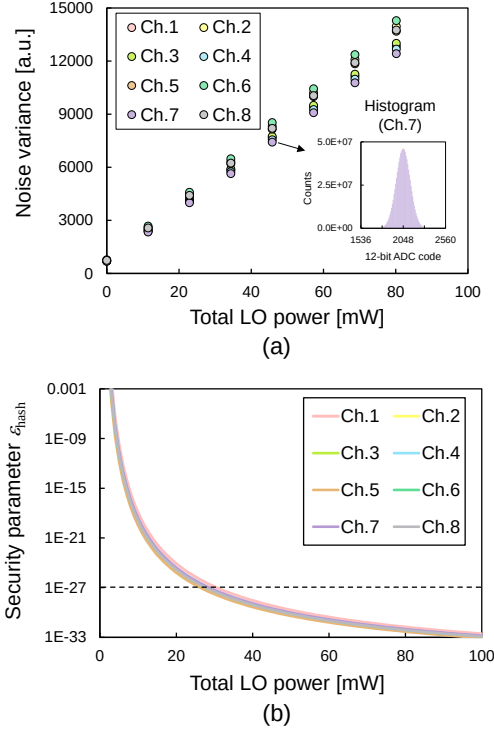


Fig. 4: Experimental results of eight-channel multiplexed operation: (a) Noise variances and (b) Security parameters ϵ_{hash} for different LO power.

channel linearly scale with the LO power ($R^2 > 0.9999$), which indicates that vacuum fluctuation or quantum noise is dominant. The non-zero plots at total LO power of 0 mW show the effect of background classical noise. The background noise was suppressed below -10 dB for all channels at total LO power of 45.8 mW corresponding to LO power of approximately 4 mW for each BPD. This result agrees well with the clearance measurements shown in Fig. 2.

We analysed the security of the 100-Gbit/s quantum random number generation. A security framework considering classical and quantum side information [15] was utilized. We calculated the lower bound of min-entropy of a single measurement outcome drawn from a random variable X conditioned on the quantum side information E , $H_{\min}(X|E)$ from the measured noise variances and QRNG setting parameters such as sampling ratio and bit resolution. The conditional min-entropy reached 8.15–8.20 bit/sample at total LO power of 45.8 mW. The min-entropy values exceed the required threshold of 6.25 bit/sample in the setup with η of 0.625 and bit resolution of 10 bits, which satisfies the basic criteria in the security framework. Then, a security parameter of hashing ϵ_{hash} was estimated from the conditional min-entropy and the size of Toeplitz matrix. Fig. 4(b) shows the security parameters for different total LO power. All eight channels achieve a low security parameter of less than 1×10^{-27} at total LO power of 40 mW with sufficient margin.

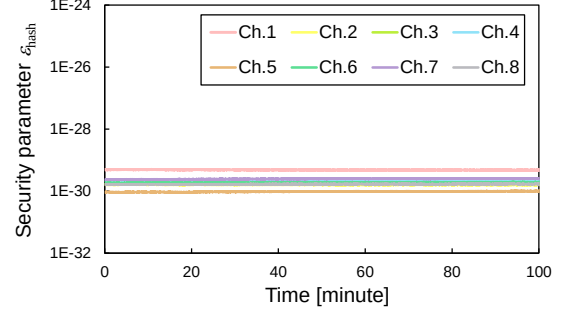


Fig. 5: Long-term monitoring results of security parameters.

Table. 1: Results of NIST SP 800-90B.

	Output 1	Output 2	Output 3	Output 4
Chi-square test	✓	✓	✓	✓
Length of LRS test	✓	✓	✓	✓
IID permutation test	✓	✓	✓	✓
Restart test	✓	✓	✓	✓

We measured security parameters for 100 minutes at total LO power of 45.8 mW. Fig. 5 shows the results. The setup achieved security parameters of less than 5×10^{-30} for all channels in a stable manner. This value corresponds to 100-year continuous secure operation with a failure probability of 3.1×10^{-13} . Typically, a failure probability of 1×10^{-9} is set as a target value, and thus the setup achieves very high security in long-term continuous operation. Finally, four digital random number outputs from SFP modules were stored on a computer and evaluated based on randomness test according to NIST SP 800-90B [17]. Table 1 summarizes the test results. All four outputs passed the test.

Conclusions

We have developed a four-channel BPD circuit board operating at low power for the measurement of vacuum fluctuations and demonstrated scaling up the parallel channels of a spatially multiplexed QRNG from four to eight. The eight-channel QRNG achieved a real-time aggregated rate of $12.5 \text{ Gbit/s} \times 8 = 100 \text{ Gbit/s}$. The security of the QRNG was evaluated based on the security framework considering classical and quantum side information. A low security parameter of less than 5×10^{-30} was achieved at total LO power of 45.8 mW for all channels. The output random numbers passed the test according to NIST SP 800-90B. The total LO power was half or one quarter of the maximum output of a typical high-power DFB laser diode, which suggests that further scaling to 200 Gbit/s with 16 channels or 400 Gbit/s with 32 channels is feasible while maintaining high security.

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